Downscaling Municipal Soybean Yield Labels to Pixel Predictions with MSTNet

Paraná, Brazil

Heitor Trielli Zanoni Ferreira

Motivation

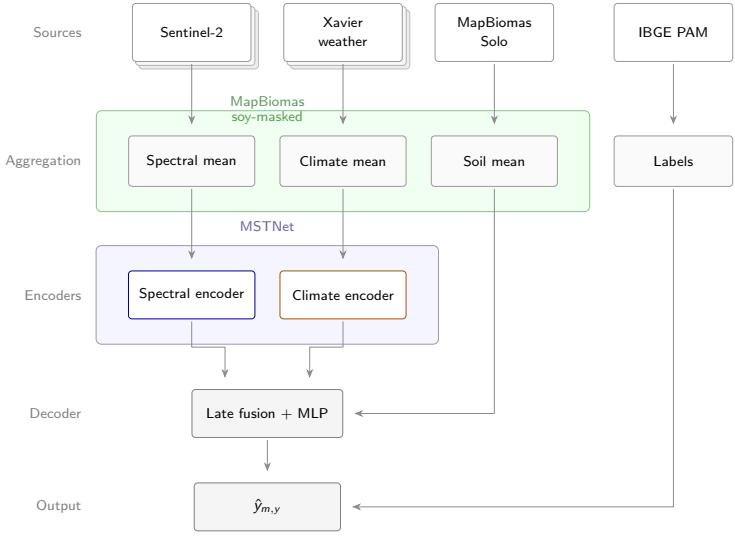
- Yield estimates matter for food security, insurance, and risk.
- At scale, the best labels we have are municipal (IBGE PAM).
- Field-level evaluation needs finer maps $\Rightarrow$ downscaling.
- Gridded Sentinel-2, climate, and soil let us learn from coarse PAM and predict per soy pixel.

Song et al. (2022): PAM$\rightarrow$pixel with RF on phenology. We use multi-frequency Transformers and year leave-one-out.

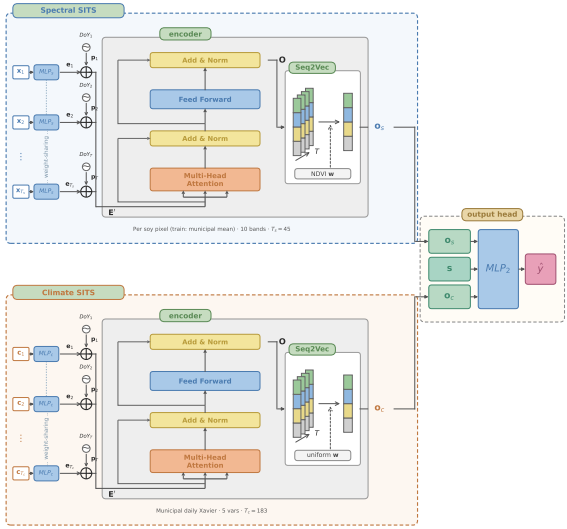
Study area and data

- Paraná soybean productivity from IBGE PAM (t/ha).
- MapBiomas soy mask (class 39) selects cells before aggregation.
- Sentinel-2: 10 L2A bands at $30\,\mathrm{m} \times 30\,\mathrm{m}$.
- Xavier climate: daily weather at $\sim 10\,\mathrm{km} \times 10\,\mathrm{km}$.
- MapBiomas Solo: clay, silt, sand, SOC at $30\,\mathrm{m} \times 30\,\mathrm{m}$ (0–30 cm).
- Fixed season window: early October through late March.
- Keep municipality–years where MapBiomas soy area is 0.5–1.5$\times$ PAM area.

Training path: municipal means



MSTNet architecture



After training: pixel scoring

- Training used municipal means; inference can use all soy pixels.
- Same weights f_θ score every soy pixel in a municipality.
- Inputs are the raw pixel values.
- Average pixel scores $\rightarrow$ municipal prediction (or any polygon).
- We could train pixel-wise, but it is too costly.

Evaluation

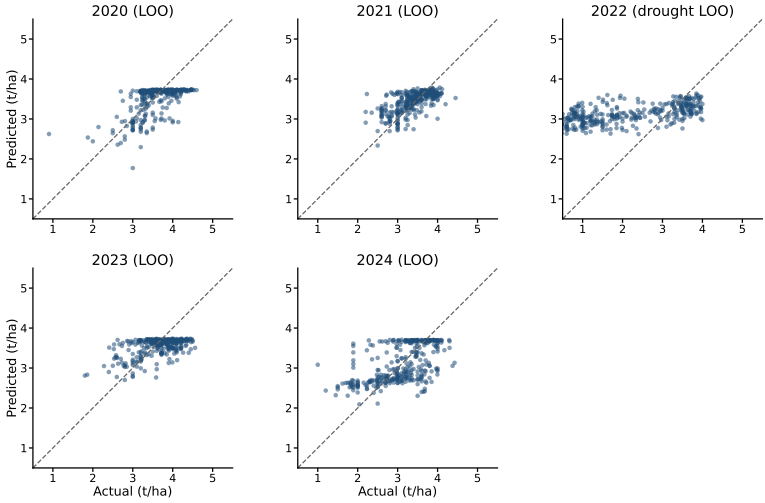
- **Year LOO** (primary)
  - Leave one harvest year out (2020–2024).
  - Train on the other four; report only the held-out year.
  - Select checkpoints on municipal-mean; also score the pixel path.
- **Spatial holdout**
  - Hold out 20% of municipalities; seasons appear in both pools.
  - Tests unseen places in known years.

Year-LOO metrics

		Municipal-mean		Pixel path		
LOO year	n	RMSE	R^2	RMSE	R^2	ΔR^2
2020	346	0.411	0.233	0.395	0.291	+0.058
2021	343	0.320	**0.410**	0.323	**0.402**	−0.008
2022	342	1.342	−0.300	1.404	−0.422	−0.122
2023	339	0.433	0.199	0.424	0.231	+0.032
2024	336	0.577	0.259	0.568	0.284	+0.025
Mean	—	0.617	0.160	0.623	0.157	−0.003
Mean excl. 2022	—	0.435	0.275	0.427	**0.302**	+0.027

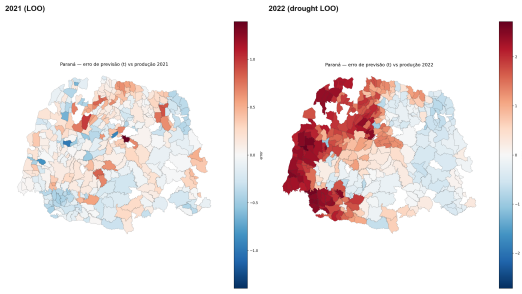
RMSE in t/ha.

Predicted vs. observed



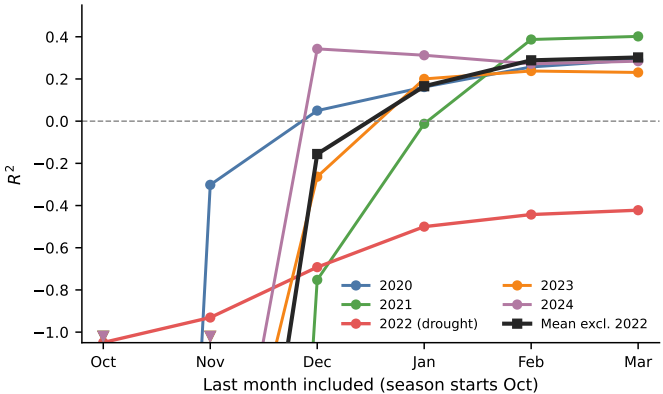
Drought year 2022

- Only drought year in the panel.
- Left out $\Rightarrow$ regime unseen in training.
- Over-predicts West / Northwest.
- OOD stress test, not a typical fold.



Best fold (2021) vs. drought fold (2022).

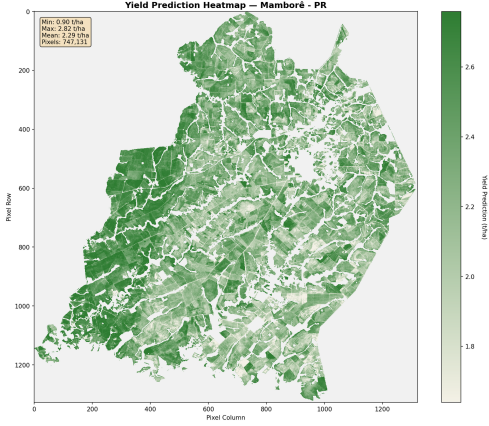
Partial-season scoring



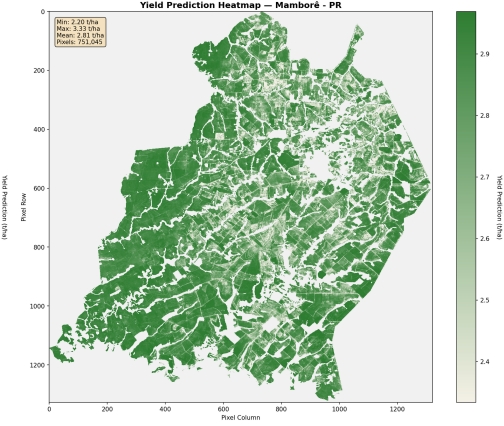
Excl. 2022: mean R^2 reaches 0.288 by February, 0.302 with full Oct–Mar.

Intramunicipal maps (Mamboré)

2021 (LOO)



2022 (drought LOO)



Qualitative downscaling—no field yields to validate against.

Modality ablations (pixel path)

Variant	Mean R^2	Excl. 2022
Full MSTNet (S2 + climate + soil)	0.157	**0.302**
No soil	0.147	0.234
No Sentinel-2	−0.027	0.040
No climate	−0.112	−0.017

- Ordinary years: climate $\gg$ Sentinel-2 $\gg$ soil.

Spatial holdout (80/20 municipalities)

Subset	n	RMSE	MAE	R^2
2020	70	0.330	0.262	0.426
2021	68	0.305	0.233	0.409
2022	69	0.694	0.468	**0.636**
2023	68	0.411	0.323	0.283
2024	68	0.385	0.300	0.581
Mean (intra-year)	—	0.425	0.317	0.467
Pooled	343	0.448	0.317	**0.736**

RMSE/MAE in t/ha. Municipal-mean path; seasons appear in both pools.

Limitations and next steps

- The model often gets the year right but ranks municipalities poorly within a year.
- Management, cultivar, and planting cycle are unobserved.
- Next: ranking / correlation losses, generative auxiliary tasks, stress-aware climate pooling, more drought years in training.

Thank you

Appendix: encoder math

Per-timestep MLP and DoY positional encoding ($i \in \{s, c\}$):

$$\mathbf{e}_{i,t} = MLP_i(\mathbf{x}_{i,t}),$$
$$\mathbf{p}_{i,t} = \left[\sin(\mathrm{DoY}_{i,t}/\tau^{2k/d}),\ \cos(\mathrm{DoY}_{i,t}/\tau^{2k/d})\right]_{k=1}^{d/2},$$
$$\mathbf{e}'_{i,t} = \mathbf{e}_{i,t} + \mathbf{p}_{i,t}, \qquad \tau = 1000.$$

Transformer on $\mathbf{E}'_i$ yields $\mathbf{O}_i$. Seq2Vec pools to one vector:

$$\mathbf{w} = \mathrm{softmax}\left([\mathrm{NDVI}_t]_{t=1}^{T_s}\right), \qquad \mathbf{o}_s = \sum_{t=1}^{T_s} w_t\, \mathbf{o}_{s,t}.$$

Climate: mean of $\mathbf{O}_c$ over valid days $\rightarrow \mathbf{o}_c$.

Appendix: decoder and loss

Late fusion:

$$\hat{y} = MLP_2\big([\mathbf{o}_s; \mathbf{o}_c; \mathbf{s}]\big).$$

Standardized municipal-mean MSE:

$$\tilde{y}_{m,y} = \frac{y_{m,y} - \mu}{\sigma}, \qquad \mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(m,y)\in\mathcal{D}} \big(\hat{y}^{\text{train}}_{m,y} - \tilde{y}_{m,y}\big)^2.$$

μ, σ from training years of the fold; metrics use $\sigma\hat{y} + \mu$.

Appendix: training paths

Train: average soy-pixel inputs $\rightarrow$ one municipal prediction $\rightarrow$ PAM

$$\bar{\mathbf{x}}_{m,y,t} = \frac{1}{|\mathcal{P}_{m,y,t}|} \sum_{p \in \mathcal{P}_{m,y,t}} \mathbf{x}_{p,t}$$

Score: predict each soy pixel $\rightarrow$ one municipal prediction (mean of $\hat{y}_p$)

$$\hat{y}_{m,y} = \frac{1}{|\mathcal{P}_{m,y}|} \sum_{p \in \mathcal{P}_{m,y}} \hat{y}_p$$